

Science — Cbse Half-Yearly Exam — 80 Marks

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	Assertion (A): Uncontrolled cell divisions leading to tumour formation are a result of errors in mitosis, rather than errors in meiosis. Reason (R): Faulty meiosis may cause early pregnancy loss or reduced fertility, while errors in mitosis lead to an abnormal number of chromosomes in body cells. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
2	What is the key difference between scalar and vector quantities? (A) Scalars are specified by just their numerical value, while vectors require specifying both direction and magnitude. (B) Scalars have both magnitude and direction, while vectors have only magnitude. (C) Scalars can only be positive, while vectors can be positive or negative. (D) Scalars are used for time, while vectors are used for distance. <i>[Ch 4: Describing Motion Around Us]</i>	1
3	When studying the motion of a falling object, what assumption is made in building a model? (A) The object's shape is perfectly spherical (B) The object falls at a constant speed (C) Gravity varies depending on the object's mass (D) Air resistance is neglected to understand the basic effect of gravity <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1



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4	On a graph paper, the scale chosen for time on the x-axis is 5 divisions = 1 s. How many divisions will represent 4 s on the x-axis? (A) 9 divisions (B) 10 divisions (C) 20 divisions (D) 15 divisions <i>[Ch 4: Describing Motion Around Us]</i>	1
5	Which tissue is responsible for the increase in length and girth of plants? (A) Connective tissue (B) Skeletal tissue (C) Meristematic tissue (D) Nervous tissue <i>[Ch 3: Tissues in Action]</i>	1
6	Assertion (A): In an oxygen molecule, the two oxygen atoms are joined by a double bond. Reason (R): Two oxygen atoms share two electrons each to complete their octet, forming two pairs of shared electrons. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 9: Atomic Foundations of Matter]</i>	1
7	In the airplane fuel incident, approximately how many litres of fuel was the aircraft short? (A) About 10,000 litres (B) About 15,000 litres (C) About 20,000 litres (D) About 22,300 litres <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
8	When a coconut is brought down at a very high velocity to hit a hard surface and stops in a very short time, what causes the shell to break? (A) A decrease in the mass of the coconut (B) A small force applied over a long time (C) A very large force exerted by the ground to change its velocity quickly (D) The gravitational force of the Earth pulling it downwards <i>[Ch 6: How Forces Affect Motion]</i>	1



Q.NO.	QUESTIONS	MARKS
9	Assertion (A): Whenever one object exerts a force on a second object, the second object simultaneously exerts an equal and opposite force on the first object. Reason (R): Newton's third law of motion applies to all types of forces, whether contact or non-contact. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 6: How Forces Affect Motion]</i>	1
10	In the process of coagulation, what does a coagulant do? (A) Makes smaller particles clump together and settle down (B) Dissolves the solute completely in the solvent (C) Spins the mixture rapidly to separate components (D) Converts a liquid directly into vapour <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
11	Mature Red Blood Cells (RBCs) in humans have a short lifespan because they lack a nucleus and cannot repair or divide. Approximately how long do they survive? (A) 7 days (B) 30 days (C) 120 days (D) 365 days <i>[Ch 2: Cell: The Building Block of Life]</i>	1
12	Raja Ramanna is often called the Father of the Indian Nuclear Programme. What did he promote? (A) The use of nuclear weapons for defence (B) The peaceful use of nuclear energy for development (C) The replacement of all power plants with nuclear plants (D) The export of nuclear technology to other nations <i>[Ch 9: Atomic Foundations of Matter]</i>	1
13	In which direction are planetary winds deflected in the Northern Hemisphere due to the Earth's rotation? (A) Towards the right (B) Towards the left (C) Upwards into the atmosphere (D) Downwards towards the surface <i>[Ch 13: Earth as a System: Energy, Matter, and Life]</i>	1



Q.NO.	QUESTIONS	MARKS
14	When an external force is removed from a deformed spring, the spring returns to its original shape. What causes this return to the original shape? (A) Internal forces undoing the deformation (B) External gravitational force (C) The change in the mass of the spring (D) The kinetic energy of the spring <i>[Ch 7: Work, Energy, and Simple Machines]</i>	1
15	Assertion (A): Tendons connect bone to bone and allow joint movement. Reason (R): Tendons are made of tough connective tissue that transmits force from muscle to bone. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 3: Tissues in Action]</i>	1
16	In 1958, F.C. Steward demonstrated that cells from the vascular phloem of carrot can regenerate whole plants. What specific components were present in the nutrient medium used in this experiment? (A) Simple sugars and hormones (B) Complex sugars and minerals (C) Simple sugars and antibiotics (D) Complex sugars and hormones <i>[Ch 3: Tissues in Action]</i>	1
17	Why do plants look wilted when they do not get enough water? (A) The lysosomes break down the cell wall. (B) The Golgi apparatus stops packaging proteins. (C) The mitochondria halt energy production. (D) The vacuole loses water, cells become less firm, and the plant wilts. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
18	Assertion (A): An equation in science is primarily a calculation tool designed for finding numerical answers. Reason (R): Mathematics in science is a powerful language for thinking that helps express compact statements about how quantities are related. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1



Q.NO.	QUESTIONS	MARKS
19	Assertion (A): We cannot hear the sound of a bell ringing in a closed jar after most of the air is pumped out. Reason (R): Sound requires a medium to travel. (i) Both A and R are true, but R is not the correct explanation of A. (ii) Both A and R are true, and R is the correct explanation of A. (iii) A is true, but R is false. (iv) A is false, but R is true. <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
20	The biological sex of a baby is determined by the chromosome contributed by: (A) The mother only (B) Both parents equally (C) The father, who contributes either X or Y (D) The mother's hormone levels <i>[Ch 11: Reproduction: How Life Continues]</i>	1
Section B Attempt 5 out of 7.		
21	At what minimum difference in boiling points can two miscible liquids be separated by distillation? <i>[Ch 5: Exploring Mixtures and their Separation]</i>	2
	OR What is the difference between a law and a theory in science? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	
22	Explain why the cell membrane model is called the 'fluid mosaic' model. <i>[Ch 2: Cell: The Building Block of Life]</i>	2
	OR An object of mass 5 kg is near the Earth's surface. Calculate the gravitational force acting on it. (Take $g = 9.8 \text{ m s}^{-2}$) <i>[Ch 6: How Forces Affect Motion]</i>	
23	Define 'rate of change' and state how it relates to average velocity. <i>[Ch 4: Describing Motion Around Us]</i>	2
24	What compounds make the bone matrix rigid, and what is the nature of the cartilage matrix? <i>[Ch 3: Tissues in Action]</i>	2
25	Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+}. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride.	2



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 9: Atomic Foundations of Matter]</i>	
Section C Attempt 5 out of 7.		
26	When creating a simple model of a cricket ball being hit for a six, on what basis do scientists decide which details to include and which to ignore? Give one example of each. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	3
	OR Define the momentum of an object. How does Newton's second law of motion apply to situations where the mass of the object is not constant? <i>[Ch 6: How Forces Affect Motion]</i>	
27	When cooking oil is mixed with water, it forms a separate layer. Which substance will form the top layer and which apparatus is used to separate these immiscible liquids? On what property is this separation based? <i>[Ch 5: Exploring Mixtures and their Separation]</i>	3
	OR Define average acceleration. How does its direction relate to the velocity when the magnitude of velocity is decreasing in a straight line motion? <i>[Ch 4: Describing Motion Around Us]</i>	
28	Differentiate between Rough Endoplasmic Reticulum (RER) and Smooth Endoplasmic Reticulum (SER) based on their structure and primary functions. <i>[Ch 2: Cell: The Building Block of Life]</i>	3
29	How does a movable pulley system differ from a fixed pulley in terms of mechanical advantage? Give one real-life application of pulley systems. <i>[Ch 7: Work, Energy, and Simple Machines]</i>	3
30	What is differentiation in plants, and what changes do cells undergo during this process? <i>[Ch 3: Tissues in Action]</i>	3
Section D Attempt 4 out of 6.		
31	Explain the force of friction and how it affects the motion of objects. When is an object able to overcome friction and start moving? A table is moved across the floor at a constant velocity using a horizontal force F . What is the magnitude of the frictional force exerted by the floor on the table? Justify your answer using Newton's laws. <i>[Ch 6: How Forces Affect Motion]</i>	5



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	OR Explain the mass by mass percentage and mass by volume percentage methods of expressing the concentration of a solution. A solution contains 5 g of glucose dissolved in water to make 100 mL of solution. Calculate its concentration using the appropriate method and justify why that method is suitable. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	
32	Explain how a microscope enables us to study cell structure. Describe the three main features that scientists have improved in microscopes over the years, and demonstrate how the estimated size of a cell and its total magnification are calculated. <i>[Ch 2: Cell: The Building Block of Life]</i>	5
	OR Describe the step-by-step procedure for plotting a position-time graph from given data. In your explanation, include how to draw the axes, choose suitable scales, plot the data points, and connect them to form the graph. <i>[Ch 4: Describing Motion Around Us]</i>	
33	Explain how lateral meristem and intercalary meristem contribute to growth in plants. Describe the location and function of each type, and give examples of everyday observations that demonstrate the action of intercalary meristem. <i>[Ch 3: Tissues in Action]</i>	5
34	Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	5
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Cell Theory and Cell Culture The Cell Theory was formulated through the work of several scientists. In 1838, Matthias Schleiden reported that all plants are made up of cells. In 1839, Theodor Schwann found that all animals are also made up of cells. In 1855, Rudolf Virchow stated that new cells are formed only from pre-existing cells. Building on the understanding of cells, scientists developed cell culture — a method to grow plant and animal cells outside the body. Cells are taken from an organism and placed in a nutrient-rich medium. The right temperature, acidic or alkaline conditions, and moisture under sterile conditions must be maintained to keep the culture safe. (i) Which scientist stated that new cells are formed only from pre-existing cells, and in which year? (ii) State any two principles of the classical Cell Theory. (iii) A laboratory technician maintains proper temperature and nutrient supply but forgets to maintain sterile conditions during cell culture. What is likely to happen and why? (iv) An Austrian botanist proposed that any living plant cell can develop into a complete plant if provided with suitable nutrients and favourable conditions. What is this special ability of plant cells called? <i>[Ch 2: Cell: The Building Block of Life]</i>	5
	OR Case Study: Plant Protective and Supporting Tissues A botanist comparing plants from different habitats observes that desert plants have a thick waxy coating on their surface, while aquatic plants float easily on water. She notices that young plant stems bend without breaking, but the hard shells of walnuts are extremely tough and cannot bend. She examines the transverse section of a sunflower stem under a microscope and identifies several types of permanent tissues. She finds that permanent tissues can be simple (composed of only one type of cell) or complex (composed of more than one type of cell), and each tissue is specialised to perform a specific function such as support, transport, or storage. (i) What is the waxy layer of cutin on the epidermis called? (ii) Which type of simple permanent tissue allows plant stems and tendrils to bend without breaking? Name the chemical responsible for this flexibility. (iii) A walnut shell is hard and strong. Identify the tissue, the chemical deposited in its cell walls, and state whether its cells are living or dead. (iv) How does the specialised parenchyma in aquatic plants help them survive in their habitat? <i>[Ch 3: Tissues in Action]</i>	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Kinematic Equations for Vehicles on a Straight Road A motorbike is moving on a straight road with an initial velocity of 28 m/s. The rider applies brakes and the motorbike decelerates uniformly, coming to rest after travelling a distance of 98 m. On a parallel track, a car starts from rest and accelerates uniformly, reaching a final velocity of 24 m/s in 6 seconds. A physics student watching both vehicles decides to compare their accelerations and the distances they cover using the kinematic equations for motion in a straight line with constant acceleration. (i) What is the average acceleration of the car during the 6 seconds? (ii) Using the equation $v^2 = u^2 + 2as$, calculate the acceleration of the motorbike. (iii) Calculate the distance travelled by the car in 6 seconds and the time taken by the motorbike to come to a stop. (iv) Compare the magnitudes of acceleration of the motorbike and the car. Which one is greater? <i>[Ch 4: Describing Motion Around Us]</i>	5
37	Case Study: Scientific Predictions and Testable Questions Rohan predicts it will rain because the clouds look dark. His friend Priya asks him to make his prediction scientifically testable. In science, predictions are not guesses—they are reasoned expectations based on evidence and careful thinking. Good scientific questions look for measurable evidence and past patterns, going beyond simple observations like 'clouds look dark.' When predictions match observations, confidence in the underlying science grows. But when they do not, scientists re-examine their assumptions, models, or measurements. Even weather forecasts can go wrong because tiny differences in conditions can grow over time, yet such failures are not a weakness of science—they are its greatest strength. (i) What is the key difference between a scientific prediction and a guess? (ii) Why are questions about measurable data and past patterns more useful for testing a prediction than simple yes/no questions? (iii) If a scientist's prediction about the weather turns out to be wrong, what steps should the scientist take, and why is this process important for science? (iv) Weather forecasts sometimes fail even though they use measurements and models. Why is this NOT considered a weakness of science? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	5

